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MEAN ESTIMATION BIAS IN LEAST SQUARES ESTIMATION OF AUTOREGRESSIVE PROCESSES*

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Estimation of the parameters of an autoregressive process with a mean that is a function of time is considered. Approximate expressions for the bias of the least squares estimator of the autoregressive parameters that is due to estimating the unknown mean function are derived. For the case of a mean function that is a polynomial in time, a reparameterization that isolates the bias is given. Using the approximate expressions, a method of modifying the least squares estimator is proposed. A Monte Carlo study of the second-order autoregressive process is presented. The Monte Carlo results agree well with the approximate theory and, generally speaking, the modified least squares estimators performed better than the least squares estimator. For the second-order process we also considered the empirical properties of the estimated generalized least squares estimator of the mean function and the error made in predicting the process one, two and three periods in the future.

1. Introduction

Let the time series Y_t satisfy

$$Y_t = \sum_{i=1}^{r} X_{ti}\beta_i + P_t, \qquad t = 1,2,\dots, \tag{1}$$

where X_{ti} is a fixed function of time for $i = 1,2,\dots,r$, P_t is a stationary pth-order normal autoregressive process satisfying

$$P_t = \sum_{j=1}^{p} \alpha_j P_{t-j} + e_t, \tag{2}$$

and $\{e_t\}$ is a sequence of normal independent $(0,\sigma^2)$ random variables. Assume that the roots of the characteristic equation,

$$m^p - \alpha_1 m^{p-1} - \cdots - \alpha_p = 0, \tag{3}$$

lie inside the unit circle. Assume that

$$\boldsymbol{D}_n(\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{D}_n = \mathrm{O}(1), \tag{4}$$

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and

$$D_n^{-1}X_t' = O(n^{-\frac{1}{2}}), \qquad t = 1,2,\dots,n,$$

where

$$X_t = (X_{t1}, X_{t2},\dots, X_{tr}), \qquad t = 1,2,\dots,n,$$

$$X = (X_1', X_2',\dots, X_n')',$$

and

$$D_n = \operatorname{diag}\left[\left\{\sum_{t=1}^{n} X_{t1}^2\right\}^{\frac{1}{2}},\dots,\left\{\sum_{t=1}^{n} X_{tr}^2\right\}^{\frac{1}{2}}\right].$$

Also assume that there exists a non-singular $r \times r$ matrix C such that

$$X_t = X_{t-m}C^m \qquad \text{for all } t, \tag{5}$$

where $C^0 = I_r$. For example, these assumptions on X_{ti} are satisfied by polynomial and trigonometric polynomial functions of time.

For the case $r = 1$, $p = 1$ and $X_t \equiv 1$, Mariott and Pope (1954), Orcutt and Winokur (1969), Salem (1971), Bora-Senta and Kounias (1980) and Lee (1981) proposed several estimators of α_1. These authors also used Monte Carlo studies to compare the small sample properties of the various estimators. See Lee (1981). The small sample properties of the least squares estimator of α have received little attention for higher-order processes. Salem (1971) extended the method of Mariott and Pope (1954) to obtain expressions for the approximate biases of the least squares estimator of α for the case $r = 1$, $p = 2$ and $X_{t1} \equiv 1$. The method proposed by Salem has no immediate extension for processes with $p > 2$. Bora-Senta and Kounias (1980) considered an iterative method of moments procedure as an alternative to the least squares estimation procedure for higher-order processes with $r = 1$ and $X_{t1} \equiv 1$. Ansley and Newbold (1980) compared the properties of the maximum likelihood predictor, the exact least squares predictor and the conditional least squares predictor of Y_{n+1}, Y_{n+2} and Y_{n+10} by means of simulation. Fuller and Hasza (1981) compared the maximum likelihood predictor with the truncated least squares predictor for some first-order autoregressive processes. The Monte Carlo results of their study indicated that the maximum likelihood predictor is only marginally superior. Kobayashi (1983) and Spitzer and Baillie (1983) considered the small sample properties of the estimators and the predictors when $r = 1$ and X_t is an autoregressive process independent of e_t. Tjøstheim and Paulsen (1983) derived formulas for the large sample bias of Yule Walker estimates for the case $r = 1$ and $X_t \equiv 1$. Walker (1984) obtained asymptotic expressions for the first three moments of the conditional least squares estimator of a pth-order autoregressive model with a constant mean.

We shall consider two commonly used procedures for estimating $\boldsymbol{\alpha}$. The first procedure is the first step of the Durbin two-step procedure [see Johnston (1972)] in which Y_t is regressed on $(\boldsymbol{X}_t, Y_{t-1}, Y_{t-2}, \ldots, Y_{t-p})$ and the coefficient of Y_{t-i} estimates α_i. There are a total of $n-p$ observations in the regression. Because the $\boldsymbol{X}_t$ satisfy (5), this procedure is equivalent to computing regression residuals $\tilde{P}_{t-i}$, $i = 0, 1, \ldots, p$, where

$$\tilde{P}_{t-i} = Y_{t-i} - \boldsymbol{X}_{t-i}\left[\sum_{j=p+1}^{n} \boldsymbol{X}_{j-i}'\boldsymbol{X}_{j-i}\right]^{-1}\left[\sum_{j=p+1}^{n} \boldsymbol{X}_{j-i}'Y_{j-i}\right],$$

and then estimating $\boldsymbol{\alpha}$ by regressing $\tilde{P}_t$ on $\tilde{P}_{t-1}, \tilde{P}_{t-2}, \ldots, \tilde{P}_{t-p}$. We denote the estimator of $\boldsymbol{\alpha}$ obtained in this way by $\tilde{\boldsymbol{\alpha}}$. See Rao (1967) for a description of this procedure for the polynomial mean function.

The second procedure consists of two steps. First, the ordinary least squares estimator $\hat{\boldsymbol{\beta}}$ of $\boldsymbol{\beta}$ is constructed, where

$$\hat{\boldsymbol{\beta}} = (\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'\boldsymbol{Y}, \tag{6}$$

and

$$\boldsymbol{Y} = (Y_1, Y_2, \ldots, Y_n)'.$$

Let the least squares residuals be denoted by $\hat{P}_t$, where

$$(\hat{P}_1, \hat{P}_2, \ldots, \hat{P}_n)' = \boldsymbol{Y} - \boldsymbol{X}\hat{\boldsymbol{\beta}} = \boldsymbol{M}\boldsymbol{Y}, \tag{7}$$

and

$$\boldsymbol{M} = \boldsymbol{I} - \boldsymbol{X}(\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'. \tag{8}$$

The second step of procedure two consists of estimating $\boldsymbol{\alpha}$ by regressing $\hat{P}_t$ on $(\hat{P}_{t-1}, \hat{P}_{t-2}, \ldots, \hat{P}_{t-p})$. We call this estimator $\hat{\boldsymbol{\alpha}}$.

In the first procedure both $\boldsymbol{\alpha}$ and $\boldsymbol{\beta}$ are estimated by one regression and hence only $n-p$ observations are used to estimate $\boldsymbol{\beta}$. In the second procedure, $\boldsymbol{\alpha}$ and $\boldsymbol{\beta}$ are estimated separately and all n observations are used in estimating $\boldsymbol{\beta}$. We establish in the next section that $\mathrm{E}[\hat{\boldsymbol{\alpha}} - \tilde{\boldsymbol{\alpha}}] = \mathrm{O}(n^{-2})$.

In this paper we obtain an approximate expression for the bias in the estimators of $\boldsymbol{\alpha}$ that arises from estimating $\boldsymbol{\beta}$. We propose an adjustment in the estimators based upon the expression for the approximate bias. Using Monte Carlo simulation, the proposed estimator is compared with the original estimator.

2. Bias in estimators of $\boldsymbol{\alpha}$

The estimator of $\boldsymbol{\alpha}$ obtained by method one can be written as

$$\tilde{\boldsymbol{\alpha}} = \tilde{\boldsymbol{H}}^{-1}\tilde{\boldsymbol{N}}, \tag{9}$$

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where

$$\tilde{\boldsymbol{H}} = (n-p)^{-1} \sum_{t=p+1}^{n} \tilde{\boldsymbol{F}}'_{t-1}\tilde{\boldsymbol{F}}_{t-1},$$

$$\tilde{\boldsymbol{N}} = (n-p)^{-1} \sum_{t=p+1}^{n} \tilde{\boldsymbol{F}}'_{t-1}\tilde{P}_t,$$

$$\tilde{\boldsymbol{F}}_{t-1} = \left(\tilde{P}_{t-1}, \tilde{P}_{t-2}, \ldots, \tilde{P}_{t-p}\right).$$

From the arguments of Fuller and Hasza (1981) and Lee (1981), it follows that $\tilde{\boldsymbol{\alpha}}$ is integrable and

$$\mathrm{E}[\tilde{\boldsymbol{\alpha}}] = \boldsymbol{\alpha} + \mathrm{E}\left[\boldsymbol{H}^{-1}\boldsymbol{A}\boldsymbol{H}^{-1}(\boldsymbol{A}\boldsymbol{\alpha} - \boldsymbol{d}) - \boldsymbol{H}^{-1}(\boldsymbol{A}\boldsymbol{\alpha} - \boldsymbol{d})\right] + \mathrm{O}(n^{-2}), \quad (10)$$

where

$$\boldsymbol{A} = \tilde{\boldsymbol{H}} - \boldsymbol{H},$$

$$\boldsymbol{d} = \tilde{\boldsymbol{N}} - \boldsymbol{N},$$

$$\boldsymbol{H} = \mathrm{E}\left[(n-p)^{-1} \sum_{t=p+1}^{n} \boldsymbol{F}'_{t-1}\boldsymbol{F}_{t-1}\right],$$

$$\boldsymbol{N} = \mathrm{E}\left[(n-p)^{-1} \sum_{t=p+1}^{n} \boldsymbol{F}'_{t-1}P_t\right],$$

$$\boldsymbol{F}_{t-1} = \left(P_{t-1}, P_{t-2}, \ldots, P_{t-p}\right).$$

The bias in $\tilde{\boldsymbol{\alpha}}$ arises from two sources. The first source of bias is inherent in estimating the product of the inverse of $\boldsymbol{H}$ and the vector $\boldsymbol{N}$. The second source of bias results from estimating the unknown function $\boldsymbol{X}_t\boldsymbol{\beta}$. The approximate bias in $\tilde{\boldsymbol{\alpha}}$ arising from estimating the mean function is given by $\mathrm{E}[-\boldsymbol{H}^{-1}(\boldsymbol{A}\boldsymbol{\alpha} - \boldsymbol{d})]$. An expression for $\mathrm{E}[\boldsymbol{A}\boldsymbol{\alpha} - \boldsymbol{d}]$ is derived in Theorem 1.

Theorem 1. Let Y_t satisfy model (1) and let $\tilde{\boldsymbol{\alpha}}$ be given by (9). Assume that the roots of eq. (3) lie inside the unit circle and that $\boldsymbol{X}_t$ satisfies conditions (4) and (5). Then the ith *coordinate of* $\mathrm{E}\{\boldsymbol{A}\boldsymbol{\alpha} - \boldsymbol{d}\}$ *is*

$$(n-p)^{-1}\mathrm{tr}\left[\left(\boldsymbol{X}'_{(-p)}\boldsymbol{X}_{(-p)}\right)^{-1}\boldsymbol{X}'_{(-p)}\boldsymbol{B}_i\boldsymbol{X}_{(-p)}\right], \quad (11)$$

where

$$\boldsymbol{B}_i = \mathrm{E}\left\{ \boldsymbol{e}_{(0)} \boldsymbol{P}'_{(-i)} \right\},$$

$$\boldsymbol{P}'_{(-i)} = \left(P_{p-i+1}, P_{p-i+2}, \ldots, P_{n-i} \right),$$

$$\boldsymbol{e}'_{(0)} = \left(e_{p+1}, e_{p+2}, \ldots, e_n \right),$$

$$\boldsymbol{X}'_{(-p)} = \left(\boldsymbol{X}'_{p+1}, \boldsymbol{X}'_{p+2}, \ldots, \boldsymbol{X}'_n \right).$$

Proof. Note that, under the first procedure,

$$\begin{aligned} \tilde{P}_{t-i} &= Y_{t-i} - \boldsymbol{X}_{t-i} \left[\sum_{j=p+1}^{n} \boldsymbol{X}'_{j-i} \boldsymbol{X}_{j-i} \right]^{-1} \sum_{j=p+1}^{n} \boldsymbol{X}'_{j-i} Y_{t-i} \\ &= Y_{t-i} - \boldsymbol{X}_t \boldsymbol{C}^{-i} \left[(\boldsymbol{C}^{-i})' \sum_{j=p+1}^{n} \boldsymbol{X}'_j \boldsymbol{X}_j \boldsymbol{C}^{-i} \right]^{-1} (\boldsymbol{C}^{-i})' \sum_{j=p+1}^{n} \boldsymbol{X}'_j Y_{t-i} \\ &= Y_{t-i} - \boldsymbol{X}_t \left[\sum_{j=p+1}^{n} \boldsymbol{X}'_j \boldsymbol{X}_j \right]^{-1} \sum_{j=p+1}^{n} \boldsymbol{X}'_j Y_{t-i}, \end{aligned}$$

and hence

$$\tilde{\boldsymbol{P}}_{(-i)} = \boldsymbol{M}_{(-p)} \boldsymbol{P}_{(-i)}, \tag{12}$$

where

$$\boldsymbol{M}_{(-p)} = \boldsymbol{I} - \boldsymbol{X}_{(-p)} \left(\boldsymbol{X}'_{(-p)} \boldsymbol{X}_{(-p)} \right)^{-1} \boldsymbol{X}'_{(-p)}.$$

Therefore, a single matrix $\boldsymbol{M}_{(-p)}$ can be used to construct all residuals. It follows that

$$\tilde{\boldsymbol{\alpha}} - \boldsymbol{\alpha} = \left(\tilde{\boldsymbol{F}}' \tilde{\boldsymbol{F}} \right)^{-1} \tilde{\boldsymbol{F}}' \boldsymbol{e}_{(0)} = \left(\boldsymbol{F}' \boldsymbol{M}_{(-p)} \boldsymbol{F} \right)^{-1} \boldsymbol{F}' \boldsymbol{M}_{(-p)} \boldsymbol{e}_{(0)},$$

where

$$\tilde{\boldsymbol{F}} = \boldsymbol{M}_{(-p)} \left(\boldsymbol{P}_{(-p)}, \boldsymbol{P}_{(-p+1)}, \ldots, \boldsymbol{P}_{(-1)} \right),$$

$$\boldsymbol{F} = \left(\boldsymbol{P}_{(-p)}, \boldsymbol{P}_{(-p+1)}, \ldots, \boldsymbol{P}_{(-1)} \right).$$

We have

$$\mathrm{E}\{\boldsymbol{A\alpha}-\boldsymbol{d}\}=(n-p)^{-1}\mathrm{E}\left\{\boldsymbol{F}'\boldsymbol{X}_{(-p)}\left(\boldsymbol{X}_{(-p)}'\boldsymbol{X}_{(-p)}\right)^{-1}\boldsymbol{X}_{(-p)}'\boldsymbol{e}_{(0)}\right\},$$

and the ith coordinate of $\mathrm{E}\{\boldsymbol{A\alpha}-\boldsymbol{d}\}$ is

$$(n-p)^{-1}\mathrm{E}\left\{\boldsymbol{P}_{(-i)}'\boldsymbol{X}_{(-p)}\left(\boldsymbol{X}_{(-p)}'\boldsymbol{X}_{(-p)}\right)^{-1}\boldsymbol{X}_{(-p)}'\boldsymbol{e}_{(0)}\right\}$$

$$=(n-p)^{-1}\mathrm{tr}\left[\left(\boldsymbol{X}_{(-p)}'\boldsymbol{X}_{(-p)}\right)^{-1}\boldsymbol{X}_{(-p)}'\boldsymbol{B}_i\boldsymbol{X}_{(-p)}\right]. \qquad \square$$

It is possible to evaluate (11) for any set of explanatory variables satisfying (5). The autoregressive process can be written as the infinite moving average

$$P_t=\sum_{j=0}^{\infty}\omega_j e_{t-j},$$

where $\omega_0=1$. It follows that the $\boldsymbol{B}_i$ of Theorem 1 is

$$\boldsymbol{B}_i=\begin{bmatrix} 0 & 0 & \cdots & 0 & 1 & \omega_1 & \omega_2 & \cdots & \omega_{n-i-p-1} \\ 0 & 0 & \cdots & 0 & 0 & 1 & \omega_1 & \cdots & \omega_{n-i-p-2} \\ \vdots & & & & & & & & \\ 0 & 0 & \cdots & 0 & 0 & 0 & 0 & \cdots & 1 \\ 0 & 0 & \cdots & 0 & 0 & 0 & 0 & \cdots & 0 \\ \vdots & & & & & & & & \\ 0 & 0 & \cdots & 0 & 0 & 0 & 0 & \cdots & 0 \end{bmatrix}\sigma^2, \qquad (13)$$

where the first i columns and the last i rows of $\boldsymbol{B}_i$ contain only zeros. The matrix $\boldsymbol{B}_i\boldsymbol{X}_{(-p)}$ can be created efficiently because ω_i satisfy the homogeneous difference equation associated with the autoregressive process. Let $\boldsymbol{Z}_t$ be the tth row of $\sigma^{-2}\boldsymbol{B}_0\boldsymbol{X}_{(-p)}$. Then

$$\boldsymbol{Z}_t=\alpha_1\boldsymbol{Z}_{t+1}+\alpha_2\boldsymbol{Z}_{t+2}+\cdots+\alpha_p\boldsymbol{Z}_{t+p}+\boldsymbol{X}_t, \qquad t=n,n-1,\ldots,$$

where $\boldsymbol{Z}_{n+1},\ldots,\boldsymbol{Z}_{n+p}$ are zero. The tth row of $\boldsymbol{B}_i\boldsymbol{X}_{(-p)}$ is $\sigma^2\boldsymbol{Z}_{t+i}$ and

$$\boldsymbol{X}_{(-p)}'\boldsymbol{B}_i\boldsymbol{X}_{(-p)}=\sum_{t=p+1}^{n}\boldsymbol{X}_t'\boldsymbol{Z}_{t+i}\sigma^2.$$

The relatively simple bias expression of Theorem 1 follows from assumption (5). The bias expression for general $\boldsymbol{X}$ is given in the appendix.

The bias for certain special cases can be evaluated explicitly. The contribution to the bias arising from the estimation of a polynomial mean function is given in Theorem 2.

Theorem 2. Assume that the conditions of Theorem 1 are satisfied and assume that

$$\boldsymbol{X}_t = (1, t, \ldots, t^{r-1}), \qquad r \geq 1.$$

Then

$$\mathrm{E}\{\boldsymbol{A\alpha} - \boldsymbol{d}\} = (n-p)^{-1} r\sigma^2 \left(\sum_{j=0}^{\infty} \omega_j \right) \boldsymbol{J} + \mathrm{O}(n^{-2})$$

$$= (n-p)^{-1} r\sigma^2 \left(1 - \sum_{j=1}^{p} \alpha_j \right)^{-1} \boldsymbol{J} + \mathrm{O}(n^{-2}), \tag{14}$$

where $\boldsymbol{J} = (1, 1, \ldots, 1)'$ *and the* ω_j *satisfy the set of homogeneous difference equations*

$$\omega_j = \alpha_1 \omega_{j-1} + \alpha_2 \omega_{j-2} + \cdots + \alpha_p \omega_{j-p}, \qquad j = 1, 2, \ldots,$$

with $\omega_0 = 1$ *and* $\omega_j = 0$ *for* $j < 0$.

Proof. It is clear that $\boldsymbol{X}_t$ satisfies the conditions (4) and (5). Using $\boldsymbol{B}_i$ of Theorem 1 given in (13) we have

$$\boldsymbol{X}'_{(-p)} \boldsymbol{B}_i \boldsymbol{X}_{(-p)} = \sum_{k=p+1}^{n-i-p} \boldsymbol{X}'_k \sum_{j=0}^{n-i-p-k} \omega_j \boldsymbol{X}_{k+i+j} \sigma^2$$

$$= \sum_{j=0}^{n-i-p-1} \omega_j \left(\sum_{k=p+1}^{n-i-j} \boldsymbol{X}'_k \boldsymbol{X}_{k+i+j} \right) \sigma^2$$

$$= \sum_{j=0}^{n-i-p-1} \omega_j \left(\sum_{k=p+1}^{n} \boldsymbol{X}'_k \boldsymbol{X}_k \right) \sigma^2$$

$$+ \sum_{j=0}^{n-i-p-1} \omega_j \left[\sum_{k=p+1}^{n} \boldsymbol{X}'_k \left(\boldsymbol{X}_{k+i+j} - \boldsymbol{X}_k \right) \right] \sigma^2$$

$$- \sum_{j=0}^{n-i-p-1} \omega_j \left(\sum_{k=n-i-j+1}^{n} \boldsymbol{X}'_k \boldsymbol{X}_{k+i+j} \right) \sigma^2.$$

Note that

$$\left(X'_{(-p)}X_{(-p)}\right)^{ls} = C_{ls}n^{-(l+s+1)} + \mathrm{O}\left(n^{-(l+s+2)}\right),$$

for some constants C_{ls}, where $(X'X)^{ls}$ is the (l, s)th element of $(X'_{(-p)}X_{(-p)})^{-1}$. Since $\sum_{j=0}^{\infty}|\omega_j|$ and $\sum_{j=0}^{\infty}j|\omega_j|$ are finite, we have

$$\operatorname{tr}\left[\left(X_{(-p)}X_{(-p)}\right)^{-1}X'_{(-p)}B_iX_{(-p)}\right] = r\sigma^2\left(\sum_{j=0}^{\infty}\omega_j\right) + \mathrm{O}(n^{-1}),$$

and the ith coordinate of $\mathrm{E}[A\alpha - d]$ is

$$(n-p)^{-1}r\sigma^2\left(\sum_{j=0}^{\infty}\omega_j\right) + \mathrm{O}(n^{-2}).$$

Because the roots of the eq. (2) are less than one in absolute value, we have $\sum_{j=0}^{\infty}\omega_j = (1 - \sum_{j=1}^{p}\alpha_j)^{-1}$. □

We now consider the bias in the second method of constructing an estimator of α.

Theorem 3. Let Y_t satisfy model (1) and let $\hat{\alpha}$ be given by

$$\hat{\alpha} = \left(\sum_{t=p+1}^{n}\hat{F}'_{t-1}\hat{F}_{t-1}\right)^{-1}\sum_{t=p+1}^{n}\hat{F}'_{t-1}\hat{P}_t = (\hat{F}'\hat{F})^{-1}\hat{F}'\hat{P}_{(0)}, \tag{15}$$

where

$$\hat{F}' = \left(\hat{F}'_{p-1}, \hat{F}'_p, \ldots, \hat{F}'_{n-1}\right),$$

$$\hat{F}_{t-1} = \left(\hat{P}_{t-1}, \hat{P}_{t-2}, \ldots, \hat{P}_{t-p}\right),$$

$$\hat{P}_t = Y_t - X_t\hat{\beta},$$

$$\hat{P}_{(0)} = \left(P_{p+1}, P_{p+2}, \ldots, P_n\right)',$$

and $\hat{\beta}$ is defined in (6). Assume that the roots of eq. (3) lie inside the unit circle and that X_t satisfies conditions (4) and (5). Then

$$\mathrm{E}\{\hat{\alpha} - \tilde{\alpha}\} = \mathrm{O}(n^{-2}).$$

Proof. We can write

$$\tilde{P}_{t(-i)} = \hat{P}_t + X_t\left(\hat{\beta} - \tilde{\beta}_{(i)}\right),$$

where $\tilde{P}_{t(-i)}$ is the tth element of the vector $\tilde{\boldsymbol{P}}_{(-i)}$ defined in (12), and

$$\tilde{\boldsymbol{\beta}}_{(i)} = \left(\boldsymbol{X}'_{(-i)}\boldsymbol{X}_{(-i)}\right)^{-1}\boldsymbol{X}'_{(-i)}\boldsymbol{Y}_{(-i)},$$

$$\boldsymbol{X}'_{(-i)} = \left(\boldsymbol{X}'_{p-i+1}, \boldsymbol{X}'_{p-i+2}, \dots, \boldsymbol{X}'_{n-i}\right).$$

It follows that

$$\begin{aligned}\boldsymbol{D}_n\left(\hat{\boldsymbol{\beta}} - \tilde{\boldsymbol{\beta}}_{(i)}\right) &= \left[\boldsymbol{D}_n(\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{D}_n\right]\boldsymbol{D}_n^{-1}\boldsymbol{X}'\boldsymbol{P} \\ &\quad - \left[\boldsymbol{D}_n\left(\boldsymbol{X}'_{(-i)}\boldsymbol{X}_{(-i)}\right)^{-1}\boldsymbol{D}_n\right]\boldsymbol{D}_n^{-1}\boldsymbol{X}'_{(-i)}\boldsymbol{P}_{(-i)} \\ &= \left[\boldsymbol{D}_n(\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{D}_n\right]\left[\boldsymbol{D}_n^{-1}\boldsymbol{X}'\boldsymbol{P} - \boldsymbol{D}_n^{-1}\boldsymbol{X}'_{(-i)}\boldsymbol{P}_{(-i)}\right] \\ &\quad + \left[\boldsymbol{D}_n\left\{(\boldsymbol{X}'\boldsymbol{X})^{-1} - \left(\boldsymbol{X}'_{(-i)}\boldsymbol{X}_{(-i)}\right)^{-1}\right\}\boldsymbol{D}_n\right] \\ &\quad \times \boldsymbol{D}_n^{-1}\boldsymbol{X}'_{(-i)}\boldsymbol{P}_{(-i)},\end{aligned}$$

where $\boldsymbol{D}_n$ was defined following (4). Observe that

$$\begin{aligned}&\boldsymbol{D}_n^{-1}\boldsymbol{X}'\boldsymbol{P} - \boldsymbol{D}_n^{-1}\boldsymbol{X}'_{(-i)}\boldsymbol{P}_{(-i)} \\ &= \boldsymbol{D}_n^{-1}\left[\sum_{t=1}^{p-i}\boldsymbol{X}_t'P_t + \sum_{t=n-i+1}^{n}\boldsymbol{X}_t'P_t\right] + \mathrm{O}_p\left(n^{-\frac{1}{2}}\right),\end{aligned}$$

and

$$\boldsymbol{D}_n\left\{(\boldsymbol{X}'\boldsymbol{X})^{-1} - \left(\boldsymbol{X}'_{(-i)}\boldsymbol{X}_{(-i)}\right)^{-1}\right\}\boldsymbol{D}_n = \mathrm{O}(n^{-1}).$$

Therefore,

$$\mathrm{E}\left\{\hat{\boldsymbol{\beta}} - \tilde{\boldsymbol{\beta}}_{(i)}\right\} = 0,$$

and

$$\mathrm{E}\left\{\boldsymbol{D}_n\left(\hat{\boldsymbol{\beta}} - \tilde{\boldsymbol{\beta}}_{(i)}\right)\left(\hat{\boldsymbol{\beta}} - \tilde{\boldsymbol{\beta}}_{(i)}\right)'\boldsymbol{D}_n\right\} = \mathrm{O}(n^{-1}).$$

Now

$$\begin{aligned}\hat{\boldsymbol{\alpha}} - \tilde{\boldsymbol{\alpha}} &= [n^{-1}\hat{\boldsymbol{F}}'\hat{\boldsymbol{F}}]^{-1}n^{-1}\hat{\boldsymbol{F}}'\hat{\boldsymbol{P}}_{(0)} - [n^{-1}\tilde{\boldsymbol{F}}'\tilde{\boldsymbol{F}}]^{-1}n^{-1}\tilde{\boldsymbol{F}}'\tilde{\boldsymbol{P}}_{(0)} \\ &= [n^{-1}\hat{\boldsymbol{F}}'\hat{\boldsymbol{F}}]^{-1}\left[n^{-1}\hat{\boldsymbol{F}}'\hat{\boldsymbol{P}}_{(0)} - n^{-1}\tilde{\boldsymbol{F}}'\tilde{\boldsymbol{P}}_{(0)}\right] \\ &\quad + \left\{[n^{-1}\hat{\boldsymbol{F}}'\hat{\boldsymbol{F}}]^{-1} - [n^{-1}\tilde{\boldsymbol{F}}'\tilde{\boldsymbol{F}}]^{-1}\right\}\left[n^{-1}\tilde{\boldsymbol{F}}'\tilde{\boldsymbol{P}}_{(0)}\right],\end{aligned}$$

where

$$\tilde{\boldsymbol{P}}_{(0)} = \left(\tilde{P}_{p+1}, \tilde{P}_{p+2}, \ldots, \tilde{P}_n\right) = \hat{\boldsymbol{P}}_{(0)}.$$

For fixed i and j,

$$\begin{aligned} n^{-1} \sum_{t=p+1}^{n} \tilde{P}_{t-i}\tilde{P}_{t-j} &= n^{-1} \sum_{t=p+1}^{n} \left\{\left[\hat{P}_{t-i} + \boldsymbol{X}_{t-i}\left(\hat{\boldsymbol{\beta}} - \tilde{\boldsymbol{\beta}}_{(i)}\right)\right]\right. \\ &\qquad \left.\times\left[\hat{P}_{t-j} + \boldsymbol{X}_{t-j}\left(\hat{\boldsymbol{\beta}} - \tilde{\boldsymbol{\beta}}_{(j)}\right)\right]\right\} \\ &= n^{-1} \sum_{t=p+1}^{n} \hat{P}_{t-i}\hat{P}_{t-j} + n^{-1} \sum_{t=p+1}^{n} \hat{P}_{t-i}\boldsymbol{X}_{t-j}\left(\hat{\boldsymbol{\beta}} - \tilde{\boldsymbol{\beta}}_{(j)}\right) \\ &\quad + n^{-1} \sum_{t=p+1}^{n} \hat{P}_{t-j}\boldsymbol{X}_{t-i}\left(\hat{\boldsymbol{\beta}} - \tilde{\boldsymbol{\beta}}_{(i)}\right) \\ &\quad + n^{-1} \sum_{t=p+1}^{n} \boldsymbol{X}_{t-i}\left(\hat{\boldsymbol{\beta}} - \tilde{\boldsymbol{\beta}}_{(i)}\right)\boldsymbol{X}_{t-j}\left(\hat{\boldsymbol{\beta}} - \tilde{\boldsymbol{\beta}}_{(j)}\right). \end{aligned}$$

Note that

$$\begin{aligned} \sum_{t=p+1}^{n} \hat{P}_{t-i}\boldsymbol{X}_{t-j}\boldsymbol{D}_n^{-1} &= \sum_{t=p+1}^{n} \hat{P}_{t-i}\boldsymbol{X}_{t-i}\boldsymbol{C}_{|j-i|}\boldsymbol{D}_n^{-1} \\ &= \sum_{t=p+1}^{n} \hat{P}_t\boldsymbol{X}_t\boldsymbol{C}_{|j-i|}\boldsymbol{D}_n^{-1} + \sum_{t=p+1-i}^{p} \hat{P}_t\boldsymbol{X}_t\boldsymbol{C}_{|j-i|}\boldsymbol{D}_n^{-1} \\ &\quad - \sum_{t=n-i+1}^{n} \hat{P}_t\boldsymbol{X}_t\boldsymbol{C}_{|j-i|}\boldsymbol{D}_n^{-1} \\ &= \sum_{t=p+1-i}^{p} \hat{P}_t\boldsymbol{X}_{t+|j-i|}\boldsymbol{D}_n^{-1} - \sum_{t=n-i+1}^{n} \hat{P}_t\boldsymbol{X}_{t+|j-i|}\boldsymbol{D}_n^{-1} \\ &= \mathrm{O}_p\left(n^{-\frac{1}{2}}\right), \end{aligned}$$

where $\boldsymbol{C}_m = \boldsymbol{C}^m$. Therefore,

$$n^{-1} \sum_{t=p+1}^{n} \tilde{P}_{t-i}\tilde{P}_{t-j} = n^{-1} \sum_{t=p+1}^{n} \hat{P}_{t-i}\hat{P}_{t-j} + \mathrm{O}_p\left(n^{-2}\right).$$

It can be shown that the expectation of powers of the elements of

$$\left[n^{-1}\sum_{t=p+1}^{n}\hat{F}_t'\hat{F}_t\right]^{-1} \quad \text{and} \quad \left[n^{-1}\sum_{t=p+1}^{n}\tilde{F}_t'\tilde{F}_t\right]^{-1}$$

are bounded. See Fuller and Hasza (1981). Therefore, since $\hat{\alpha}$ and $\tilde{\alpha}$ are square integrable we have

$$\mathrm{E}[\hat{\alpha}-\tilde{\alpha}] = \mathrm{O}(n^{-2}). \qquad \square$$

Note that for X_t given in (13), the coordinates of $\mathrm{E}[A\alpha - d]$ are all equal to $(n-p)^{-1}r\sigma^2(1-\sum_{j=1}^{p}\alpha_j)^{-1} + \mathrm{O}(n^{-2})$. Pantula (1982) obtained an analogous expression for $X_t = (t^{i_1}, t^{i_2}, \ldots, t^{i_r})$, where $0 \le i_1 < i_2 \cdots < i_r$ are integers. Theorem 2 also suggests that it is possible to isolate the effect of estimating the mean function by transforming the model (1). For $p \ge 2$, consider the following reparameterization:

$$P_t = \delta_1 P_{t-1} + \sum_{j=2}^{p}\delta_j\left(P_{t-j+1} - P_{t-j}\right) + e_t, \tag{16}$$

where

$$\alpha_1 = \delta_1 + \delta_2, \quad \alpha_j = \delta_{j+1} - \delta_j, \qquad j = 2,3,\ldots,p-1, \tag{17}$$

and $\alpha_p = -\delta_p$. Therefore, $\delta_1 = \sum_{j=1}^{p}\alpha_j$ and δ_1 is equal to one if and only if one of the roots of the eq. (3) is one. The least squares estimator $\hat{\delta}$ of δ is obtained by regressing $\hat{P}_t$ on $\hat{P}_{t-1}$, $(\hat{P}_{t-1}-\hat{P}_{t-2}), (\hat{P}_{t-2}-\hat{P}_{t-3}), \ldots, (\hat{P}_{t-p+1}-\hat{P}_{t-p})$, where $\hat{P}_t = Y_t - X_t\hat{\beta}$ and $\hat{\beta}$ is given by (6). From Theorem 2, it follows that an approximate expression for the bias in $\hat{\delta}$ arising from estimating the mean function is

$$G^{-1}f + \mathrm{O}(n^{-2}), \tag{18}$$

where

$$f = r\sigma^2(n-p)^{-1}(1-\delta_1)^{-1}(1,0,0,\ldots,0)',$$

$$G = \mathrm{E}\left[(n-p)^{-1}\sum_{t=p+1}^{n}G_t'G_t\right],$$

$$G_t = \left(P_{t-1}, P_{t-1}-P_{t-2}, P_{t-2}-P_{t-3}, \ldots, P_{t-p+1}-P_{t-p}\right).$$

The expression in (18) suggests the following method of correcting for the bias in $\hat{\delta}$ due to estimating the mean function:

(a) Obtain the least squares estimator of δ,

$$\hat{\delta} = \hat{G}^{-1}\hat{g},$$

where

$$\hat{\boldsymbol{G}} = (n-p)^{-1} \sum_{t=p+1}^{n} \hat{\boldsymbol{G}}_t' \hat{\boldsymbol{G}}_t,$$

$$\hat{\boldsymbol{G}}_t = \left(\hat{P}_{t-1}, \hat{P}_{t-1} - \hat{P}_{t-2}, \ldots, \hat{P}_{t-p+1} - \hat{P}_{t-p}\right),$$

$$\hat{\boldsymbol{g}} = (n-p)^{-1} \sum_{t=p+1}^{n} \hat{\boldsymbol{G}}_t' \hat{P}_t.$$

Modify the estimator to incorporate the condition that $\delta_1 \le 1$, where the modified estimator is

$$\boldsymbol{\delta}^* = \hat{\boldsymbol{\delta}} + \hat{\boldsymbol{G}}^{-1}\hat{\boldsymbol{h}},$$

with

$$\begin{aligned} \hat{\boldsymbol{h}} &= (\hat{h}_1, 0, \ldots, 0)' && \text{if} \quad p \ge 2, \\ &= \hat{h}_1 && \text{if} \quad p = 1, \\ \hat{h}_1 &= (\hat{G}^{11})^{-1}(1 - \hat{\delta}_1) && \text{if} \quad \hat{\delta}_1 \ge 1, \\ &= 0 && \text{if} \quad \hat{\delta}_1 < 1, \end{aligned}$$

and $\hat{G}^{11}$ is the upper left element of $\hat{\boldsymbol{G}}^{-1}$. When $\hat{\delta}_1$ is greater than unity, the elements of $\hat{\boldsymbol{\delta}}$ are modified to obtain a $\boldsymbol{\delta}^*$ such that the first element is unity.

(b) Using (18), we obtain the adjusted least squares estimator as

$$\boldsymbol{\delta}^+ = \hat{\boldsymbol{\delta}} + \hat{\boldsymbol{G}}^{-1}\hat{\boldsymbol{f}},$$

where

$$\begin{aligned} \hat{\boldsymbol{f}} &= (\hat{f}_1, 0, 0, \ldots, 0)' && \text{if} \quad p \ge 2, \\ &= \hat{f}_1 && \text{if} \quad p = 1, \\ \hat{f}_1 &= (\hat{G}^{11})^{-1}(1 - \hat{\delta}_1) && \text{if} \quad \hat{\delta}_1 \ge 1 \quad \text{or} \\ & && \text{if} \quad \hat{\sigma}^2 \hat{G}^{11} \ge (1 - \hat{\delta}_1)^2 (n-p), \\ &= \left[(n-p)(1 - \hat{\delta}_1)\right]^{-1} r\hat{\sigma}^2 && \text{otherwise,} \end{aligned}$$

and $\hat{\sigma}^2$ is the residual mean square error of the regression in (a). If $\hat{\delta}_1$ is less than unity and we use $[(n-1)(1-\hat{\delta}_1)]^{-1} r\hat{\sigma}^2$ as the first element of $\hat{\boldsymbol{f}}$, then the bias adjusted estimator of δ_1 will be greater than unity whenever $\hat{\delta}^2\hat{G}^{11}$ is greater than $(1-\hat{\delta}_1)^2(n-p)$. The element $\hat{f}_1$ is defined such that δ_1^+ is always less than or equal to one.

(c) Obtain the corresponding estimators, $\boldsymbol{\alpha}^*$ and $\boldsymbol{\alpha}^+$ using the relations in (17).

The estimator $\boldsymbol{\alpha}^+$ of $\boldsymbol{\alpha}$ is relatively easy to construct and, hence, is of practical importance. A Monte Carlo study of the estimators $\boldsymbol{\alpha}^*$ and $\boldsymbol{\alpha}^+$ is presented in the next section.

3. Monte Carlo study

Estimators for a second-order autoregressive process with mean function linear in time are studied. Samples of size 25 and 50 were generated for 12 sets of parameter values (α_1, α_2). The (α_1, α_2) and the roots of the characteristic equation are given in table 1. Independent normal (0, 1) random variables were generated for the e_t using the IMSL subroutine GGNML. The subroutine first generates uniform deviates using the method suggested by Lewis et al. (1969) and transforms the uniform deviates to normal deviates using the inverse normal routine suggested by Strecok (1968).

For the processes with one of the roots equal to one, the initial observations Y_1 and Y_2 were set equal to e_1 and $e_2+\alpha_1 e_1$, respectively. For the remaining parameter values, Y_1 and Y_2 are defined by

$$Y_1 = \{\gamma(0)\}^{\frac{1}{2}} e_1,$$

$$Y_2 = \{\gamma(0)\}^{\frac{1}{2}}\left[\alpha_1(1-\alpha_2)^{-1}e_1 + \left\{1-\alpha_1^2(1-\alpha_2)^{-2}\right\}e_2\right],$$

Table 1

The roots of the characteristic equation and the parameter values.

Set no.	Roots		Values of the parameters		
	m_1	m_2	α_1	$\alpha_2=-\delta_2$	$\delta_1=\alpha_1+\alpha_2$
1	1.0	0.0	1.0	0.00	1.00
2	0.8	0.5	1.3	−0.40	0.90
3	0.8	0.1	0.9	−0.08	0.82
4	0.8	−0.5	0.3	0.40	0.70
5	0.8	−0.8	0.0	0.64	0.64
6	0.5	0.0	0.5	0.00	0.50
7	0.5	−0.5	0.0	0.25	0.25
8	0.5	−0.8	−0.3	0.40	0.10
9	0.1	−0.5	−0.4	0.05	−0.35
10	0.1	−0.8	−0.7	0.08	−0.62
11	−0.5	−0.8	−1.3	−0.40	−1.70
12	0.7	$\pm 0.7i$	1.4	−0.98	0.42

where

$$\gamma(0) = [(1+\alpha_2)(1-\alpha_1-\alpha_2)(1+\alpha_1-\alpha_2)]^{-1}(1-\alpha_2).$$

The remaining observations are given by

$$Y_t = \alpha_1 Y_{t-1} + \alpha_2 Y_{t-2} + e_t.$$

Without loss of generality we take $\beta_0 = \beta_1 = 0$.

For each (α_1, α_2, n) combination, various point estimates are computed using the same set of observations. This is repeated for 1,000 sets of observations. Sample biases, variances and mean square errors for each estimator are obtained by averaging over the 1,000 replications. The Monte Carlo study was conducted on the National Advanced Systems (NAS) AS16. Method two is used to estimate the parameters throughout the study. For a second-order autoregressive process with the roots of its characteristic equation less than unity in absolute value,

$$H^{-1}\mathrm{E}[A\boldsymbol{\alpha} - \boldsymbol{d}] = 2(1+\alpha_2)(n-2)^{-1}(1,1)' + \mathrm{O}(n^{-2}). \tag{19}$$

The theory in section 2 is developed for stationary autoregressive error sequences. However, we include one non-stationary case in our study to estimate the performance of our procedure for boundary cases. Recall the reparameterization considered in (16). Let

$$\delta_1 = \alpha_1 + \alpha_2, \qquad \delta_2 = -\alpha_2,$$

and hence

$$\alpha_1 = \delta_1 + \delta_2, \qquad \alpha_2 = -\delta_2.$$

We present the results for the estimates of δ_1 and α_2 $(= -\delta_2)$.

Table 2 contains the empirical bias of the estimators of α_2 and δ_1 for the model with an estimated time trend. Except when both roots are negative or complex, δ_1^+ has smaller absolute bias than δ_1^*. The differences in the biases of the estimators α_2^+ and α_2^* and of α_1^+ and α_1^* are very close to $2(n-2)^{-1} \times (1+\alpha_2)$ and $4(n-2)^{-1}(1+\alpha_2)$, respectively. Therefore, for positive values of α_2 the reduction in the bias of the least squares estimator is large. When α_2 is negative, α_2^+ has larger absolute bias than α_2^*. When the roots are complex, δ_1^* underestimates the true value of δ_1, whereas δ_1^+ overestimates the true value.

The standard errors for the biases in table 3 are computed using the formula

$$\text{s.e.}(\overline{\text{bias}}) = \left[(1000)^{-1}(999)^{-1}\sum_{l=1}^{1000}\left(\text{bias}_l - \overline{\text{bias}}\right)^2\right]^{\frac{1}{2}}, \tag{20}$$

where bias_l is the bias of an estimator for the lth sample and $\overline{\text{bias}}$ is the sample mean reported in table 2. For $n = 25$, the standard errors of the estimated bias of α_2^* and α_2^+ are about 0.006 when the roots are real. The standard errors of the estimated bias of δ_1^* and δ_1^+ range from 0.005 to 0.014. The standard errors of the means of α_2^* and α_2^+ for $n = 50$, are about 0.004 and those of the means of δ_1^* and δ_1^+ range from 0.001 to 0.009.

Table 4 contains the mean square errors of the estimators for the time trend model. For positive values of δ_1, δ_1^+ has smaller mean square error than δ_1^*. When α_2 is negative, the mean square error of δ_1^+ is larger than that of δ_1^*. For values of δ_1 greater than 0.5, the mean square error of δ_1^+ is about one-half of that of δ_1^*. Generally, α_2^+ has smaller mean square error than α_2^*. For negative α_2 of large absolute value, α_2^* has smaller mean square error than α_2^+. Generally speaking, the adjustment produces large gains in mean square error for those parametric configurations where the original estimators have large mean square errors and produces losses for those parameters where the original

Table 2

Empirical bias of estimators for the time trend model.

Set no.	True α_2	Bias of the estimators α_2^*	α_2^+	δ_1^*	δ_1^+
		$n = 25$			
1	0.00	−0.163	−0.078	−0.442	−0.269
2	−0.40	−0.062	−0.006	−0.237	−0.119
3	−0.08	−0.149	−0.068	−0.313	−0.148
4	0.40	−0.251	−0.131	−0.427	−0.186
5	0.64	−0.283	−0.141	−0.483	−0.200
6	0.00	−0.134	−0.043	−0.269	−0.084
7	0.25	−0.193	−0.081	−0.315	−0.091
8	0.40	−0.201	−0.074	−0.312	−0.060
9	0.05	−0.130	−0.033	−0.197	−0.003
10	0.08	−0.112	−0.001	−0.158	0.045
11	−0.40	0.069	0.140	0.192	0.333
12	−0.98	0.027	0.032	−0.003	0.008
		$n = 50$			
1	0.00	−0.076	−0.035	−0.211	−0.129
2	−0.40	−0.020	0.007	−0.092	−0.037
3	−0.08	−0.058	−0.019	−0.126	−0.046
4	0.40	−0.119	−0.061	−0.188	−0.072
5	0.64	−0.135	−0.066	−0.222	−0.084
6	0.00	−0.069	−0.026	−0.118	−0.033
7	0.25	−0.094	−0.042	−0.152	−0.047
8	0.40	−0.098	−0.038	−0.153	−0.034
9	0.05	−0.065	−0.020	−0.098	−0.008
10	0.08	−0.062	−0.015	−0.088	0.005
11	−0.40	−0.001	0.026	0.022	0.076
12	−0.98	0.018	0.020	0.001	0.005

estimators have small mean square errors. Therefore, the mean square function for the adjusted estimator is much flatter than the corresponding function for the least squares estimator.

Table 5 contains the standard errors for the estimated mean square errors of the estimators for the time trend model. For $n = 25$, the standard errors for the mean square errors of α_2^* and α_2^+ range from 0.0004 to 0.0050, and for $n = 50$, they range from 0.0002 to 0.0018. The standard errors for the estimated mean square errors of δ_1^* and δ_1^+ range from 0.0001 to 0.0150 for $n = 25$, and from 0.0001 to 0.0048 for $= 50$.

For the second-order autoregressive process with mean linear in time, we considered the estimated generalized least squares (EGLS) of β_0 and β_1. Note that the generalized least squares estimator of $\boldsymbol{\beta} = (\beta_0, \beta_1)'$ is

$$\hat{\boldsymbol{\beta}}_G = (\boldsymbol{X}'\boldsymbol{V}^{-1}\boldsymbol{X})^{-1}\boldsymbol{X}'\boldsymbol{V}^{-1}\boldsymbol{Y}, \tag{21}$$

Table 3

Standard errors (multiplied by 100) of the bias of estimators for the time trend model.

Set no.	True α_2	Standard errors (multiplied by 100) α_2^*	α_2^+	δ_1^*	δ_1^+
		$n = 25$			
1	0.00	0.596	0.651	0.683	0.706
2	−0.40	0.541	0.588	0.512	0.522
3	−0.08	0.572	0.623	0.713	0.750
4	0.40	0.634	0.698	0.944	1.027
5	0.64	0.647	0.716	1.075	1.184
6	0.00	0.602	0.665	0.789	0.858
7	0.25	0.636	0.706	1.028	1.134
8	0.40	0.634	0.705	1.153	1.277
9	0.05	0.654	0.725	1.125	1.244
10	0.08	0.643	0.712	1.201	1.328
11	−0.40	0.649	0.715	1.255	1.382
12	−0.98	0.212	0.226	0.157	0.164
		$n = 50$			
1	0.00	0.431	0.444	0.382	0.383
2	−0.40	0.385	0.402	0.258	0.267
3	−0.08	0.430	0.450	0.375	0.393
4	0.40	0.453	0.476	0.589	0.619
5	0.64	0.421	0.444	0.670	0.707
6	0.00	0.440	0.460	0.516	0.537
7	0.25	0.448	0.469	0.672	0.704
8	0.40	0.447	0.468	0.795	0.832
9	0.05	0.463	0.484	0.790	0.826
10	0.08	0.457	0.478	0.860	0.899
11	−0.40	0.410	0.428	0.804	0.839
12	−0.98	0.143	0.148	0.097	0.101

where $\boldsymbol{X}$ and $\boldsymbol{Y}$ are defined in section 1, and $\boldsymbol{V}$ is the variance covariance matrix of $(P_1, P_2, \ldots, P_n)$. Let $\boldsymbol{\beta}^*$ and $\boldsymbol{\beta}^+$ be the EGLS estimators of $\boldsymbol{\beta}$ obtained by estimating $\boldsymbol{V}$ using $\boldsymbol{\alpha}^*$ and $\boldsymbol{\alpha}^+$, respectively.

The empirical results of $\boldsymbol{\beta}^*$ and $\boldsymbol{\beta}^+$, for $n = 25$, are summarized in table 6. Because the errors are symmetric, the estimated generalized least squares estimator of (β_0, β_1) is unbiased. In no case did the Monte Carlo bias exceed twice the estimated standard error of the bias. The adjustment again produces large gains in mean square errors for those parametric configurations where the original estimators have large mean square errors and produces losses for those parameters where the original estimators have small mean square errors.

We also considered the empirical properties of predictors constructed from the adjusted and modified estimators of $\boldsymbol{\beta}$ and $\boldsymbol{\alpha}$. Note that

$$\begin{aligned} Y_{n+1} &= (\beta_0 + \beta_1 n)(1 - \alpha_1 - \alpha_2) + \beta_1(1 + \alpha_2) + \alpha_1 Y_n + \alpha_2 Y_{n-1} + e_{n+1}, \\ Y_{n+2} &= (\beta_0 + \beta_1 n)\left(1 - \alpha_1^2 - \alpha_2 - \alpha_1\alpha_2\right) + \beta_1(2 + \alpha_1\alpha_2) \\ &\quad + \left(\alpha_1^2 + \alpha_2\right)Y_n + \alpha_1\alpha_2 Y_{n-1} + e_{n+2} + \alpha_1 e_{n+1}, \end{aligned} \tag{22}$$

Table 4

Empirical mean square error (multiplied by 10) of estimators for the time trend model.

Set no.	True α_2	Mean square error (multiplied by 10) α_2^*	α_2^+	δ_1^*	δ_1^+
		$n = 25$			
1	0.00	0.620	0.485	2.422	1.221
2	−0.40	0.332	0.347	0.823	0.415
3	−0.08	0.547	0.434	1.488	0.783
4	0.40	1.033	0.658	2.718	1.399
5	0.64	1.220	0.712	3.492	1.803
6	0.00	0.543	0.461	1.345	0.808
7	0.25	0.776	0.563	2.048	1.368
8	0.40	0.804	0.552	2.302	1.666
9	0.05	0.598	0.537	1.655	1.548
10	0.08	0.538	0.508	1.692	1.784
11	−0.40	0.469	0.707	1.944	3.016
12	−0.98	0.052	0.061	0.025	0.028
		$n = 50$			
1	0.00	0.244	0.209	0.590	0.312
2	−0.40	0.152	0.162	0.151	0.085
3	−0.08	0.219	0.206	0.299	0.175
4	0.40	0.348	0.264	0.701	0.434
5	0.64	0.360	0.241	0.942	0.571
6	0.00	0.241	0.218	0.405	0.299
7	0.25	0.290	0.237	0.682	0.517
8	0.40	0.295	0.234	0.866	0.703
9	0.05	0.257	0.238	0.720	0.683
10	0.08	0.246	0.230	0.818	0.809
11	−0.40	0.168	0.190	0.650	0.762
12	−0.98	0.024	0.026	0.009	0.010

Table 5

Standard errors (multiplied by 100) of the mean square error of estimators for the time trend model.

Set no.	True α_2	Standard errors (multiplied by 100) α_2^*	α_2^+	δ_1^*	δ_1^+
		$n = 25$			
1	0.00	0.235	0.196	0.788	0.602
2	−0.40	0.131	0.150	0.367	0.257
3	−0.08	0.217	0.189	0.682	0.506
4	0.40	0.381	0.292	1.102	0.797
5	0.64	0.477	0.368	1.498	1.124
6	0.00	0.213	0.189	0.636	0.453
7	0.25	0.314	0.249	0.924	0.658
8	0.40	0.337	0.259	1.108	0.831
9	0.05	0.250	0.229	0.823	0.720
10	0.08	0.226	0.218	0.762	0.754
11	−0.40	0.235	0.354	0.957	1.425
12	−0.98	0.049	0.060	0.016	0.020
		$n = 50$			
1	0.00	0.102	0.092	0.237	0.187
2	−0.40	0.070	0.077	0.077	0.054
3	−0.08	0.096	0.089	0.152	0.107
4	0.40	0.167	0.138	0.352	0.258
5	0.64	0.173	0.137	0.474	0.349
6	0.00	0.107	0.100	0.215	0.164
7	0.25	0.125	0.105	0.328	0.255
8	0.40	0.130	0.105	0.423	0.338
9	0.05	0.108	0.101	0.319	0.290
10	0.08	0.109	0.104	0.366	0.359
11	−0.40	0.075	0.089	0.299	0.362
12	−0.98	0.023	0.024	0.006	0.007

Table 6

Empirical mean square error of 'EGLS' estimators for β_0 and β_1.

		$n = 25$				$n = 50$			
		Mean square error		Mean square error × 1000		Mean square error		Mean square error × 1000	
Set no.	α_2	β_0^*	β_0^+	β_1^*	β_1^+	β_0^*	β_0^+	β_1^*	β_1^+
1	0.00	2.906	2.148	49.179	47.197	4.263	2.879	22.191	21.588
2	−0.40	8.996	8.673	34.695	32.801	5.666	5.835	6.198	6.502
3	−0.08	2.913	2.973	11.474	11.195	1.760	1.791	1.912	1.976
4	0.40	1.049	1.054	4.269	4.247	0.674	0.693	0.755	0.773
5	0.64	0.777	0.767	3.175	3.092	0.474	0.478	0.518	0.527
6	0.00	0.579	0.583	2.540	2.611	0.311	0.313	0.365	0.372
7	0.25	0.274	0.280	1.172	1.207	0.134	0.135	0.160	0.161
8	0.40	0.187	0.196	0.792	0.853	0.100	0.100	0.113	0.114
9	0.05	0.101	0.102	0.461	0.468	0.047	0.047	0.054	0.054
10	0.08	0.074	0.075	0.350	0.356	0.032	0.032	0.038	0.038
11	−0.40	0.030	0.030	0.136	0.139	0.013	0.013	0.014	0.014
12	−0.98	2.490	2.394	9.643	9.238	0.570	0.557	0.594	0.584

$$Y_{n+3} = (\beta_0 + \beta_1 n)(1 - \alpha_1^3 - 2\alpha_1\alpha_2 - \alpha_1^2\alpha_2 - \alpha_2^2) + \beta_1(3 + \alpha_1^2\alpha_2 + \alpha_2^2)$$

$$+ (\alpha_1^3 + 2\alpha_1\alpha_2) Y_n + (\alpha_1^2\alpha_2 + \alpha_2^2) Y_{n-1} + e_{n+3}$$

$$+ \alpha_1 e_{n+2} + (\alpha_1^2 + \alpha_2) e_{n+1}.$$

Let Y_{n+s}^* and Y_{n+s}^+, $s = 1,2,3$, be the forecasts obtained from (22) by setting the errors equal to zero and replacing $(\boldsymbol{\beta}, \boldsymbol{\alpha})$ with $(\boldsymbol{\beta}^*, \boldsymbol{\alpha}^*)$ and $(\boldsymbol{\beta}^+, \boldsymbol{\alpha}^+)$, respectively. We can assume, without loss of generality, that $\boldsymbol{\beta} = \mathbf{0}$. Using the arguments of Fuller and Hasza (1981), it can be shown that the expected value of the prediction error is zero. Note that

$$\mathrm{E}\left[(Y_{n+1} - Y_{n+1}^*)^2\right] = 1 + \mathrm{E}\Big[\big\{(\alpha_1 - \alpha_1^*) Y_n + (\alpha_2 - \alpha_2^*) Y_{n-1}$$

$$- (1 - \alpha_1^* - \alpha_2^*)(\beta_0^* + \beta_1^* n) - (1 + \alpha_2^*)\beta_1^*\big\}^2\Big].$$

Therefore, for example, it is only necessary to simulate the distribution of

$$(\alpha_1 - \alpha_1^*) Y_n + (\alpha_2 - \alpha_2^*) Y_{n-1}$$

$$- (1 - \alpha_1^* - \alpha_2^*)(\beta_0^* + \beta_1^* n) - (1 + \alpha_2^*)\beta_1^*,$$

to obtain an estimate of the mean square error of the one-period prediction error. Similar expressions hold for two- and three-period predictions and for $(\boldsymbol{\beta}^+, \boldsymbol{\alpha}^+)$.

Table 7 contains the Monte Carlo variances (mean square errors) of the error made in predicting the second-order process one, two, and three periods in the future. The large sample theory gives 1, $1 + \alpha_1^2$ and $1 + \alpha_1^2 + (\alpha_1^2 + \alpha_2)^2$ as the mean square error of one-, two-, and three-period predictions, respectively. Generally, the mean square errors of the prediction errors of Y_{n+s}^+ are smaller than those of Y_{n+s}^*. For $n = 50$, the mean square errors of the two procedures are close to each other. The estimated standard errors for the mean square errors of Y_{n+s}^* when $n = 25$ range from 0.012 to 0.024, from 0.010 to 0.089 and from 0.010 to 0.037, for one-, two-, and three-period predictions, respectively. The standard errors for the Y_{n+s}^+ predictions are smaller than those of Y_{n+s}^*. The estimated standard errors for $n = 50$ were about 0.007, 0.011, 0.036 for one-, two-, and three-period predictions, respectively.

Table 7

Empirical mean square errors of various prediction errors.

		$n = 25$						$n = 50$						$n = \infty$		
	α_2	Y^*_{n+1}	Y^+_{n+1}	Y^*_{n+2}	Y^+_{n+2}	Y^*_{n+3}	Y^+_{n+3}	Y^*_{n+1}	Y^+_{n+1}	Y^*_{n+2}	Y^+_{n+2}	Y^*_{n+3}	Y^+_{n+3}	Y_{n+1}	Y_{n+2}	Y_{n+3}
1	0.00	1.34	1.18	3.01	2.51	4.69	3.93	1.14	1.08	2.48	2.23	3.93	3.46	1.00	2.00	3.00
2	−0.40	1.38	1.26	4.54	3.97	8.41	7.29	1.15	1.13	3.44	3.32	6.06	5.83	1.00	2.69	4.35
3	−0.08	1.35	1.25	2.90	2.61	4.15	3.77	1.14	1.13	2.27	2.22	3.15	3.09	1.00	1.81	2.34
4	0.40	1.34	1.25	1.53	1.45	2.20	1.96	1.14	1.13	1.28	1.26	1.66	1.66	1.00	1.09	1.33
5	0.64	1.35	1.28	1.25	1.22	2.04	2.01	1.15	1.13	1.12	1.11	1.71	1.72	1.00	1.00	1.41
6	0.00	1.32	1.28	1.81	1.79	1.97	2.02	1.13	1.13	1.50	1.50	1.62	1.64	1.00	1.25	1.31
7	0.15	1.28	1.26	1.24	1.24	1.37	1.40	1.13	1.13	1.11	1.11	1.21	1.22	1.00	1.00	1.06
8	0.40	1.26	1.27	1.26	1.27	1.67	1.72	1.14	1.14	1.16	1.16	1.52	1.53	1.00	1.09	1.33
9	0.05	1.26	1.28	1.34	1.33	1.79	1.80	1.13	1.13	1.23	1.23	1.68	1.68	1.00	1.16	1.60
10	0.08	1.31	1.32	1.68	1.66	2.22	2.21	1.14	1.14	1.55	1.55	1.96	1.96	1.00	1.49	1.81
11	−0.40	1.34	1.38	3.13	3.14	5.38	5.38	1.13	1.13	2.87	2.87	4.78	4.77	1.00	2.69	4.35
12	−0.98	1.41	1.39	5.59	5.45	9.85	9.54	1.17	1.17	3.93	3.93	5.83	5.83	1.00	2.96	3.92

4. Summary

An expression for the bias in the estimators of the parameters of an autoregressive process that is due to estimating the mean function is given. For an autoregressive process with mean function a polynomial in time, the bias in the least squares estimator arising from estimating the mean function has a relatively simple form. This form can be used to correct the estimator for the bias arising from estimating the mean function. For a second-order autoregressive process with mean function polynomial in time, the biases due to estimating the unknown mean function in the least squares estimators of α_2 and δ_1 are proportional to $(n-2)^{-1}(1+\alpha_2)+\mathrm{O}(n^{-2})$. The Monte Carlo study demonstrates that the adjusted estimators behave well with respect to bias and mean square error, compared to the least squares estimators, when $\delta_1(=\alpha_1+\alpha_2)$ is positive and α_2 is positive. The Monte Carlo study also demonstrates that the mean square error of the adjusted estimator is smaller than that of the least squares estimator for a wide range of parameter values. The mean square error function for the adjusted estimator is a more nearly constant function of the parameters than is the mean square error function of the least squares estimator. The estimated generalized least squares estimator of $\boldsymbol{\beta}$ computed using the adjusted estimator $\boldsymbol{\alpha}^+$ has flatter mean square error function than the estimated generalized least squares estimator of $\boldsymbol{\beta}$ computed using the least squares estimator $\boldsymbol{\alpha}^*$. The prediction errors of the predictions using $\boldsymbol{\beta}^+$ and $\boldsymbol{\alpha}$ had smaller empirical mean square errors than those using $\boldsymbol{\beta}^*$ and $\boldsymbol{\alpha}^*$ for most parameters. Generally, predictions based on $\boldsymbol{\alpha}^+$ were superior to those based on $\boldsymbol{\alpha}^*$ for parameters where $\boldsymbol{\alpha}^+$ was superior to $\boldsymbol{\alpha}^*$.

Appendix

In this appendix we evaluate the leading term in the bias of the estimator of $\boldsymbol{\alpha}$ that arises from estimating the mean function. Let $\boldsymbol{H}_i$ be the $(n-p)\times n$ matrix

$$\boldsymbol{H}_i=\mathrm{E}\{\boldsymbol{P}_{(-i)}\boldsymbol{P}'\}$$

$$=\begin{pmatrix} \gamma(p-i) & \gamma(p-i-1) & \cdots & \gamma(n-i-2) & \gamma(n-i-1) \\ \vdots & \vdots & & \vdots & \vdots \\ \gamma(n-i-1) & \gamma(n-i-2) & \cdots & \gamma(i-1) & \gamma(i) \end{pmatrix},$$

where

$$\boldsymbol{P}_{(-i)}=(P_{p-i+1},P_{p-i+2},\ldots,P_{n-i})',$$

$$\boldsymbol{P}'=(P_1,P_2,\ldots,P_n).$$

Let T be the $n \times n$ matrix

$$T = \begin{bmatrix} 1 & 0 & 0 & \cdots & 0 \\ \omega_1 & 1 & 0 & \cdots & 0 \\ \omega_2 & \omega_1 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & & \vdots \\ \omega_{n-1} & \omega_{n-2} & \omega_{n-3} & \cdots & 1 \end{bmatrix},$$

and let $T_{(i)}$ be the $(n-p)\times n$ matrix created by deleting the first i and last $p-i$ rows of T.

For general x-variables, the leading term in the bias for the ith coefficient that arises from estimating the mean function is

$$\begin{aligned} &\mathrm{E}\{\tilde{P}'_{(-i)}[\tilde{P}_{(0)} - \alpha_1 \tilde{P}_{(-1)} - \cdots - \alpha_p \tilde{P}_{(-p)}]\} \\ &= \mathrm{E}\{\tilde{P}'_{(-i)}[e_{(0)} - (X_{(0)} - \alpha_1 X_{(-1)} - \cdots - \alpha_p X_{(-p)})(\hat{\beta} - \beta)]\} \\ &= \mathrm{E}\{[P_{(-i)} - X_{(-i)}(X'X)^{-1}X'P]'[e_{(0)} - W_{(0)}(X'X)^{-1}X'P] \\ &= -\operatorname{tr} X'_{(-i)}T_{(p)}X(X'X)^{-1}\sigma^2 + \operatorname{tr} X'VX(X'X)^{-1}X'_{(-i)}W_{(0)}(X'X)^{-1} \\ &\quad - \operatorname{tr} X'H_i'W_{(0)}(X'X)^{-1}, \end{aligned} \tag{A.1}$$

where

$$W_{(0)} = X_{(0)} - \alpha_1 X_{(-1)} - \cdots - \alpha_p X_{(-p)} \quad \text{and} \quad V = \mathrm{E}\{PP'\}.$$

Let Z_t be the variables defined below (13) of the text. Then the expression (A.1) is approximately

$$\begin{aligned} &-\operatorname{tr} X'_{(-i)}Z_{(p)}(X'X)^{-1}\sigma^2 + \operatorname{tr} Z'Z(X'X)^{-1}X'_{(-i)}W_{(0)}(X'X)^{-1}\sigma^2 \\ &-\operatorname{tr} Z'_{(0)}T'_{(i)}W_{(0)}(X'X)^{-1}\sigma^2, \end{aligned}$$

where

$$Z_{(p)} = T'_{(p)}X,$$

because

$$V \doteq TT'\sigma^2, \qquad H_i \doteq T_{(i)}T'\sigma^2.$$

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